derham

De Rham complexes and FEEC diagrams in LaTeX

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Abstract

derham builds de Rham complexes and finite element exterior calculus (FEEC) diagrams from mathematical labels. It uses tikz-cd for layout and provides independent control of spaces, differential operators, and inter-row maps.

LPPL 1.3c or later. Version 0.3.0 adds cross-degree row connections, arbitrary positive dimensions for generic and form notation, and local styling. See `CHANGELOG.md` for the release history.

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1 Installation and five-minute start

Copy `derham.sty` beside your document or install it in a local TEX tree. The package requires a current LaTeX kernel, amsmath, and tikz-cd. Then write:

```
\usepackage{derham}
...
\[\derhamcomplex\]
\[\derhamcommuting\]
```

The two commands render, respectively:

$$V^0 \xrightarrow{d^0} V^1 \xrightarrow{d^1} V^2 \xrightarrow{d^2} V^3$$

$$\begin{array}{ccccccc} V^0 & \xrightarrow{d^0} & V^1 & \xrightarrow{d^1} & V^2 & \xrightarrow{d^2} & V^3 \\ \downarrow{\scriptstyle \pi_h^0} & & \downarrow{\scriptstyle \pi_h^1} & & \downarrow{\scriptstyle \pi_h^2} & & \downarrow{\scriptstyle \pi_h^3} \\ V_h^0 & \xrightarrow{d_h^0} & V_h^1 & \xrightarrow{d_h^1} & V_h^2 & \xrightarrow{d_h^2} & V_h^3 \end{array}$$

Both commands produce a math object and belong in math mode. They work inside `\[...\]` and `equation`; do not wrap them in another `tikzcd` environment. A one-row discrete complex is obtained with `discrete=true`.

2 Notation and mathematical meaning

The default is the generic sequence $V^0 \xrightarrow{d^0} V^1 \xrightarrow{d^1} \cdots \xrightarrow{d^{n-1}} V^n$. Its discrete spaces are V_h^k with arrows d_h^k; vertical maps have labels π_h^k. These symbols are labels, not definitions of spaces or maps. Drawing a square does not establish that it commutes, nor does adding zero objects establish exactness.

Choose `notation=forms` with `spaces=sobolev` to label the continuous row $H\Lambda^k(\Omega)$.

With `notation=vector` and `spaces=sobolev`, the conventions in dimensions 1, 2, and 3 are:

$$H^1(\Omega) \xrightarrow{\frac{\mathrm{d}}{\mathrm{d}x}} L^2(\Omega)$$

$$H^1(\Omega) \xrightarrow{\mathrm{grad}} H(\mathrm{curl};\Omega) \xrightarrow{\mathrm{curl}} L^2(\Omega)$$

$$H^1(\Omega) \xrightarrow{\mathrm{grad}} H(\mathrm{curl};\Omega) \xrightarrow{\mathrm{curl}} H(\mathrm{div};\Omega) \xrightarrow{\mathrm{div}} L^2(\Omega)$$

The two-dimensional curl in this preset is scalar valued. If your vector proxy or boundary conditions use a different convention, supply explicit labels. Discrete spaces default to generic V_h^k even when the continuous row uses Sobolev names, because the precise finite element spaces depend on the discretization.

3 Continuous and discrete complexes

```
\[\derhamcomplex[dimension=3,notation=vector,spaces=sobolev]\]
```

$$H^1(\Omega) \xrightarrow{\text{grad}} H(\text{curl};\Omega) \xrightarrow{\text{curl}} H(\text{div};\Omega) \xrightarrow{\text{div}} L^2(\Omega)$$

```
\[\derhamcomplex[dimension=2,discrete=true]\]
```

$$V_h^0 \xrightarrow{d_h^0} V_h^1 \xrightarrow{d_h^1} V_h^2$$

Choose a consecutive degree interval with `from` and `to`; the defaults are 0 and `dimension`. Optional zero objects are visual endpoints only:

```
\[\derhamcomplex[from=1,to=3,zero-start=true,zero-end=true]\]
```

$$0 \longrightarrow V^1 \xrightarrow{d^1} V^2 \xrightarrow{d^2} V^3 \longrightarrow 0$$

4 Commuting diagrams

```
\[\derhamcommuting[dimension=2,notation=forms,spaces=sobolev]\]
```

$$\begin{array}{ccccc} H\Lambda^0(\Omega) & \xrightarrow{d^0} & H\Lambda^1(\Omega) & \xrightarrow{d^1} & H\Lambda^2(\Omega) \\ \downarrow{\scriptstyle \pi_h^0} & & \downarrow{\scriptstyle \pi_h^1} & & \downarrow{\scriptstyle \pi_h^2} \\ V_h^0 & \xrightarrow{d_h^0} & V_h^1 & \xrightarrow{d_h^1} & V_h^2 \end{array}$$

Use `vertical-direction=up` when the maps point from the discrete row to the continuous row. The `projection` key changes the base symbol of automatically indexed labels. For arbitrary labels, use `maps` or individual `map-`*`k`* options.

```
\[\derhamcommuting[
  dimension=2,
  vertical-direction=up,
  maps={I^0,I^1,I^2}
]\]
```

$$\begin{array}{ccccc} V^0 & \xrightarrow{d^0} & V^1 & \xrightarrow{d^1} & V^2 \\ \uparrow{\scriptstyle I^0} & & \uparrow{\scriptstyle I^1} & & \uparrow{\scriptstyle I^2} \\ V_h^0 & \xrightarrow{d_h^0} & V_h^1 & \xrightarrow{d_h^1} & V_h^2 \end{array}$$

Individual arrows can be reversed with `map-direction-`*`k`*`=up` or `down`. A mixed-direction diagram may have a different mathematical interpretation from a commuting projection diagram; the author must specify the intended identities.

5 Custom spaces and operators

List options use braces around the entire comma-separated list. An item containing a comma must itself be braced. Lists cover all degrees $0, \dots, n$, even if a range is shown. `operators` and `discrete-operators` each contain n items; `maps` contains $n + 1$.

```
\[\derhamcommuting[
  dimension=2,
  custom-spaces={X^0,X^1,X^2},
  custom-discrete-spaces={X_h^0,X_h^1,X_h^2},
  operators={D_0,D_1},
  discrete-operators={D_{h,0},D_{h,1}},
  maps={P_0,P_1,P_2}
]\]
```

$$\begin{array}{ccccc} X^0 & \xrightarrow{D_0} & X^1 & \xrightarrow{D_1} & X^2 \\ \downarrow{\scriptstyle P_0} & & \downarrow{\scriptstyle P_1} & & \downarrow{\scriptstyle P_2} \\ X_h^0 & \xrightarrow{D_{h,0}} & X_h^1 & \xrightarrow{D_{h,1}} & X_h^2 \end{array}$$

The indexed keys `space-0` through `space-3`, `discrete-space-0` through `discrete-space-3`, `operator-0` through `operator-2`, `discrete-operator-0` through `discrete-operator-2`, and `map-0` through `map-3` replace one label. An indexed override takes priority over a list, which takes priority over a preset. Indices outside the selected dimension produce an error.

For a plasma-oriented paper, spaces and operators might be named by the particular compatible spline discretization. The package does not choose those spaces:

```
\[\derhamcommuting[
  dimension=3,
  custom-discrete-spaces={V_h^0,V_h^1,V_h^2,V_h^3},
  discrete-operators={\nabla_h,\nabla_h\times,\nabla_h\cdot},
  maps={\Pi_h^0,\Pi_h^1,\Pi_h^2,\Pi_h^3}
]\]
```

$$\begin{array}{ccccccc} V^0 & \xrightarrow{d^0} & V^1 & \xrightarrow{d^1} & V^2 & \xrightarrow{d^2} & V^3 \\ \downarrow{\scriptstyle \Pi_h^0} & & \downarrow{\scriptstyle \Pi_h^1} & & \downarrow{\scriptstyle \Pi_h^2} & & \downarrow{\scriptstyle \Pi_h^3} \\ V_h^0 & \xrightarrow{\nabla_h} & V_h^1 & \xrightarrow{\nabla_h\times} & V_h^2 & \xrightarrow{\nabla_h\cdot} & V_h^3 \end{array}$$

This is an illustrative labeling, not a reproduction of a specific published diagram.

6 Mapped domains

For a map $F : \widehat{\Omega} \to \Omega$, pullback has the contravariant direction $F^* : H\Lambda^k(\Omega) \to H\Lambda^k(\widehat{\Omega})$. The default preset draws physical spaces above reference spaces and points the pullback arrows down. The exterior derivative and pullback satisfy $F^*d = dF^*$ when the relevant spaces and map meet the required hypotheses; the package only displays the labels.

```
\[\derhammapped[dimension=2,notation=forms,spaces=sobolev]\]
```

$$\begin{array}{ccccc} H\Lambda^0(\Omega) & \xrightarrow{d^0} & H\Lambda^1(\Omega) & \xrightarrow{d^1} & H\Lambda^2(\Omega) \\ \downarrow{\scriptstyle F^*} & & \downarrow{\scriptstyle F^*} & & \downarrow{\scriptstyle F^*} \\ H\Lambda^0(\widehat{\Omega}) & \xrightarrow{\widehat{d}^0} & H\Lambda^1(\widehat{\Omega}) & \xrightarrow{\widehat{d}^1} & H\Lambda^2(\widehat{\Omega}) \end{array}$$

Use `geometry-direction=physical-to-reference` when the geometric map points from Ω to $\widehat{\Omega}$. The pullback then points up. Set `pullback` to its actual symbol, for example `pullback=G^*` for $G : \Omega \to \widehat{\Omega}$. `reference-spaces` and `reference-operators` accept full lists; `maps` and individual `map-k` keys replace pullback labels.

7 Coefficient spaces

The coefficient preset places discrete function spaces above coefficient spaces. Its default coefficient objects are $\mathbb{R}^{N_k}$, horizontal maps are G, C, D in three dimensions, and vertical maps are labeled DoF^k.

```
\[\derhamcoefficients[dimension=3]\]
```

$$\begin{array}{ccccccc} V_h^0 & \xrightarrow{d_h^0} & V_h^1 & \xrightarrow{d_h^1} & V_h^2 & \xrightarrow{d_h^2} & V_h^3 \\ \downarrow{\scriptstyle \mathrm{DoF}^0} & & \downarrow{\scriptstyle \mathrm{DoF}^1} & & \downarrow{\scriptstyle \mathrm{DoF}^2} & & \downarrow{\scriptstyle \mathrm{DoF}^3} \\ \mathbb{R}^{N_0} & \xrightarrow{G} & \mathbb{R}^{N_1} & \xrightarrow{C} & \mathbb{R}^{N_2} & \xrightarrow{D} & \mathbb{R}^{N_3} \end{array}$$

The labels G, C, D denote user-chosen matrices. They do not prescribe a basis or imply identity mass matrices. Use `coefficient-spaces`, `coefficient-operators`, and `maps` for full lists; `dof` changes the base symbol for indexed vertical maps. A reconstruction diagram can use `vertical-direction=up` with appropriate reconstruction labels.

8 Named multilevel diagrams

The `derhamdiagram` environment works in math mode. Declare named rows and connect them by name. Connections may precede row declarations; row IDs must be unique. An omitted `degree` connects all corresponding columns, while `degree=1` connects only degree 1.

```
\[
\begin{derhamdiagram}[dimension=2]
  \derhamconnect[from=physical,to=reference,operator=F^*]
  \derhamconnect[from=reference,to=discrete,
                 operator=\pi_h]
  \derhamrow[name=physical,kind=continuous]
  \derhamrow[name=reference,kind=reference]
  \derhamrow[name=discrete,kind=discrete]
\end{derhamdiagram}
\]
```

$$\begin{array}{ccccc} V^0 & \xrightarrow{d^0} & V^1 & \xrightarrow{d^1} & V^2 \\ \downarrow{\scriptstyle F^*} & & \downarrow{\scriptstyle F^*} & & \downarrow{\scriptstyle F^*} \\ \widehat{V}^0 & \xrightarrow{\widehat{d}^0} & \widehat{V}^1 & \xrightarrow{\widehat{d}^1} & \widehat{V}^2 \\ \downarrow{\scriptstyle \pi_h} & & \downarrow{\scriptstyle \pi_h} & & \downarrow{\scriptstyle \pi_h} \\ V_h^0 & \xrightarrow{d_h^0} & V_h^1 & \xrightarrow{d_h^1} & V_h^2 \end{array}$$

Each `\derhamrow` accepts `name`, `kind`, `spaces`, and `operators`. Row kinds include `continuous`, `discrete`, `reference`, `coefficients`, `conforming`, `dual`, and `broken`. Row lists have $n+1$ spaces and n operators.

Each `\derhamconnect` accepts `from`, `to`, `operator`, and optional `degree`. The operator is repeated for an all-degree connection. For distinct labels by degree, declare individual connections. Set both `from-degree` and `to-degree` to connect different columns:

```
\[\begin{derhamdiagram}[dimension=2,row-sep=huge]
  \derhamrow[name=A]
  \derhamrow[name=B]
  \derhamconnect[from=A,to=B,from-degree=0,
                 to-degree=1,operator=T]
\end{derhamdiagram}\]
```

$$\begin{array}{ccccc} V^0 & \xrightarrow{d^0} & V^1 & \xrightarrow{d^1} & V^2 \\ & \searrow^{T} & & & \\ V^0 & \xrightarrow{d^0} & V^1 & \xrightarrow{d^1} & V^2 \end{array}$$

Connections between different columns are rendered as diagonal arrows. Their mathematical meaning is supplied by the author.

9 Higher dimensions and styling

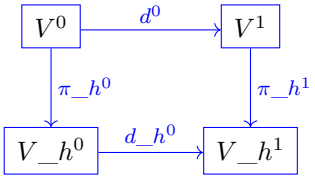
Generic and form notation accept any positive finite dimension. Indexed overrides work at every degree. Vector notation has built-in names only in dimensions 1–3. Wider diagrams may need smaller spacing or a landscape page.

```
\[\derhamcomplex[dimension=4,notation=forms,
                 space-4=X^4,operator-3=D_3,
                 column-sep=normal]\]
```

$$V^0 \xrightarrow{d^0} V^1 \xrightarrow{d^1} V^2 \xrightarrow{d^2} V^3 \xrightarrow{D_3} X^4$$

Use `arrow-style` for labeled-arrow options. Use `diagram-style` for diagram-level tikz-cd options; `font` and `baseline` are shortcuts. Settings on one command are scoped to that diagram.

```
\[\derhamcommuting[dimension=1,arrow-style=blue,
  font=\small,diagram-style={cells={nodes={draw=blue}}}]\]
```



10 Primal, dual, and broken rows

`\derhamdual` is an algebraic dual layout: V_h^k has rightward d_h^k arrows, $(V_h^k)^*$ has leftward $\widetilde{d}_h^k$ arrows, and the vertical labels are M_k. These are editable labels. The weak arrows and mass maps must be defined consistently by the author; this diagram does not assert an adjoint identity.

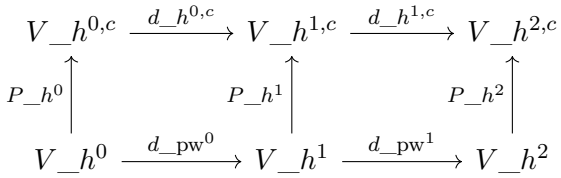
```
\[\derhamdual[dimension=2]\]
```

$$\begin{array}{ccccc} V_h^0 & \xrightarrow{d_h^0} & V_h^1 & \xrightarrow{d_h^1} & V_h^2 \\ \downarrow{\scriptstyle M_0} & & \downarrow{\scriptstyle M_1} & & \downarrow{\scriptstyle M_2} \\ (V_h^0)^* & \xleftarrow[\widetilde{d}_h^0]{} & (V_h^1)^* & \xleftarrow[\widetilde{d}_h^1]{} & (V_h^2)^* \end{array}$$

Use `dual-spaces`, `dual-operators`, `duality`, and `maps` to adopt a specific indexing and mass-map convention. The dual row can also be used in a named diagram.

`\derhambroken` places conforming $V_h^{k,c}$ spaces above broken V_h^k spaces and draws P_h^k upward. The broken row uses patchwise derivative labels d_{pw}^k; the conforming row uses $d_h^{k,c}$. These are labels for the structural relation in the cited paper. In an actual broken FEEC construction, specify the precise patchwise derivatives and projected strong operators.

```
\[\derhambroken[dimension=2]\]
```



Set `broken-spaces`, `broken-operators`, `conforming-projection`, or `maps` for the conventions in your paper. The broken and dual layouts follow the structural distinction between strong and weak sequences discussed by Güçlü, Campos Pinto, and Hadjout; they are intentionally generic rather than copies of a particular published diagram.

11 Configuration reference

`\derhamsetup{...}` sets defaults in the current TeX scope. Options on a diagram command take priority and are local to that command.

Key	Meaning and default
`dimension`	Positive integer; default 3. Built-in vector labels support 1–3.
`notation`	`generic`, `forms`, or `vector`; default `generic`.
`spaces`	`generic` or `sobolev`; default `generic`.
`discrete`	Boolean, default `false`; applies to `\derhamcomplex`.
`from`, `to`	Inclusive degree range; defaults 0 and the dimension.
`zero-start`, `zero-end`	Boolean visual endpoints; default `false`.
`custom-spaces`, `custom-discrete-spaces`	Comma lists of $n+1$ math labels.
`operators`, `discrete-operators`	Comma lists of n math labels.
`maps`	Comma list of $n+1$ vertical map labels.
`projection`	Base symbol for indexed vertical labels; default π_h.
`vertical-direction`	`down` or `up`; default `down`.
`row-sep`, `column-sep`	tikz-cd separation values; default `large`.
`diagram-style`, `arrow-style`	Scoped tikz-cd diagram and labeled-arrow options.
`font`, `baseline`	Scoped diagram font and baseline settings.
`space-`*k*, `discrete-space-`*k*	Override the degree-k space.
`operator-`*k*, `discrete-operator-`*k*	Override the arrow from degree k to $k+1$.
`map-`*k*, `map-direction-`*k*	Override a vertical map label or its direction.
`geometry-direction`, `pullback`	Geometric map orientation and pullback label for `\derhammapped`.
`reference-spaces`, `reference-operators`	Full lists for the reference row.
`coefficient-spaces`, `coefficient-operators`, `dof`	Coefficient row and degree-of-freedom labels.
`dual-spaces`, `dual-operators`, `duality`	Algebraic dual row and vertical labels.
`broken-spaces`, `broken-operators`, `conforming-projection`	Broken row and conforming projection labels.

For example, `\derhamsetup{dimension=2,column-sep=huge}` changes later diagrams in the current group. Standard tikz-cd values such as `small`, `normal`, `large`, and `huge` are accepted for spacing.

12 Published-notation regression example

The following labels follow the conforming FEEC projection relation in Güçlü, Campos Pinto, and Hadjout (2023), Eq. (10). The corresponding source is `examples/plasma-feec.tex`. Commutativity is a hypothesis on the chosen projections, not a property checked by derham.

```
\[\derhamcommuting[
  dimension=3,
  custom-discrete-spaces={V_h^{0,c},V_h^{1,c},
                          V_h^{2,c},V_h^{3,c}},
  operators={\operatorname{grad},\operatorname{curl},
             \operatorname{div}},
  discrete-operators={\operatorname{grad}_h^c,
                      \operatorname{curl}_h^c,
                      \operatorname{div}_h^c},
  maps={\Pi_h^{0,c},\Pi_h^{1,c},\Pi_h^{2,c},\Pi_h^{3,c}}
]\]
```

$$\begin{array}{ccccccc} V^0 & \xrightarrow{\text{grad}} & V^1 & \xrightarrow{\text{curl}} & V^2 & \xrightarrow{\text{div}} & V^3 \\ \downarrow{\scriptstyle \Pi_h^{0,c}} & & \downarrow{\scriptstyle \Pi_h^{1,c}} & & \downarrow{\scriptstyle \Pi_h^{2,c}} & & \downarrow{\scriptstyle \Pi_h^{3,c}} \\ V_h^{0,c} & \xrightarrow{\text{grad}_h^c} & V_h^{1,c} & \xrightarrow{\text{curl}_h^c} & V_h^{2,c} & \xrightarrow{\text{div}_h^c} & V_h^{3,c} \end{array}$$

13 Limits and roadmap

Version 0.3 supports any positive dimension in generic and form notation, with built-in vector labels in dimensions 1–3. Named rows may be stacked and connected across degrees. Publication-specific mapped multipatch presets remain future work. No externalization or automatic sizing is provided. A wide diagram may need a smaller `column-sep`, a landscape page, or a paper-specific layout. Adapted source examples are in `examples/paper-gallery.tex`.

The mapped pullback convention follows the naturality of exterior differentiation described in Arnold's FEEC notes. The distinction between conforming and broken spaces, primal and dual sequences, and patchwise versus projected operators follows Güçlü, Campos Pinto, and Hadjout, Journal of Scientific Computing (2023). These sources motivate the examples; users must select operators and spaces for their own setting.